Nathan Espejo

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Education

Western University — London, ON
Bachelor of Engineering Science in Software Engineering, Dean's Honour List (2024, 2025) — *Aug 2024 - May 2028*

Experience

VP of Data/ML (promoted from Software Developer) — Sep 2025 - Aug 2026
3D Western — *London, ON*

- Led two developers through tickets and PR review to build the club's print-approval dataset and benchmark.
- Built a labelled dataset from 4,127 scraped 3D models, rendered in Blender and class-checked by a VLM.
- Benchmarked base vision models on the clean safe, unsafe and grey classes to set the fine-tuning baseline.

Software Engineering Intern — May 2026 - Aug 2026
TeraGo — *Markham, ON*

- Automated outage ticketing for non-standard provider emails: a local LLM (Ollama) extracts the key details and the pipeline opens Jira issues through Microsoft Graph.
- Wrote Channel Finder, a Python tool that finds clear RF channels in MapInfo data, validated on 760 test cases.
- Deployed a Copilot Studio agent that answers engineers' RF questions from SharePoint documents.
- Proposed a self-hosted LLM platform (LibreChat with SharePoint and Jira) to cut Copilot licensing costs.

Projects

Campus | *Python, FastAPI, SQLite, React, TypeScript, Docker* — Aug 2026 - Present

- Built a self-hosted AI study assistant that syncs my LMS courses into SQLite with Playwright and runs on any OpenAI-compatible endpoint, local or hosted.
- Designed hybrid retrieval over 4,687 chunks from 215 course files with embedding search, cross-encoder reranking and a lexical SQL prefilter. Fixed exact phrases that scored below the semantic cutoff.
- Wrote a 20-tool agent harness with an audit log for every AI write and a bounded tool loop, with 308 tests in CI.
- Use it across six courses for deadline lookups, studying from course slides, and reviewing drafts against briefs.

Homelab & NixOS | *NixOS, Hermes Agent, Docker, Tailscale* — May 2026 - Present

- Defined a laptop and home server in one git-tracked NixOS config, with 27 hardened containers behind Tailscale.
- Run one Hermes Agent profile per host that makes changes as config edits, across 472 sessions since June 2026.
- Gated agent changes behind my approval and a nixos-safe-rebuild guard, with rollback to the last good build.
- Grew a git-versioned library of 191 agent skills run on low-cost models picked per task, pruned weekly by a curator that keeps backups of removed skills.

AWARE (Hack the 6ix, solo) | *Python, YOLOv8, YAMNet, ONNX, Qualcomm QNN* — Jul 2026

- Built an offline home-awareness device running YOLOv8 and YAMNet on an $80 Arduino UNO Q with QNN.
- Used an on-device LLM to compile commands like "when glass breaks after 10pm alert me" into deterministic rules, checked every 500 ms with no LLM in the loop.

Hackathon Awards

8 awards in 22 hackathons: NSBEHacks 3rd place, 1st for Cohere (Careerly) | Hack Western Best AI Application with Cloudflare (Auralis) | TerraHacks Best Use of MongoDB (Lumen) | Spark 2026 3rd (CareBridge, contributor)

Technical Skills

AI: LLM agents, tool calling, RAG, embeddings and reranking, MCP, VLMs, model evaluation, llama.cpp, Ollama, PyTorch, YOLOv8, ONNX
Languages: Python, TypeScript, C#, C++, SQL, Bash
Web & Infra: React, FastAPI, Next.js, NixOS, Linux, Docker, Tailscale, Caddy, Git, GitHub Actions
Tools: Hermes Agent, Microsoft Graph, Copilot Studio, Jira, SharePoint, Blender